

Fakturama Image-to-Cash — Design, in brief

Ahmed Islam Farouk Abbas · TJM Labs take-home, Part 1 · full version in `DESIGN.md`

In	one order image
Out	a saved, verified Order and its linked Invoice
Forbidden	hardcoded coordinates; any assumption about window size, theme or DPI
The hard part	not clicking — deciding whether what I am looking at is the thing I meant, and stopping when I cannot tell

Shape: `extract` → `flow` → `driver`. `extract` is pure, so it ships and is tested before a Windows VM exists. `flow` is the spec transcribed, step numbers in the comments. `driver` is the only module that knows what a window is.

Finding controls

Eclipse/SWT: UIA sees a real control tree, but almost nothing carries an `AutomationId`, names are locale-derived, and the two icons beside `Addresses` — where the spec insists on the *upper* one — have no name at all.

TIER	METHOD	BUYS
T1	scoped search — resolve the window, then search inside it	"Save" means <i>this Order's</i> Save
T2	anchor-relative walk — a named landmark, then the nth child	"the upper icon" as tree position, not a pixel
T3	tooltip / description / enabled state	separates identical siblings
T4	screenshot the anchor's own UIA rectangle , ask a model which	one unnamed icon cannot sink the run

T2 is the answer to "no hardcoded coordinates." Ordering is a property of the widget hierarchy, read from the live tree at call time. 104 selectors in one dict.

Reading the image

JOB	TOOL	WHY
What the table says	LLM vision, JSON schema	returns structure — which column is <code>Disc.</code> , which value is <code>BILLING</code>
Where the cells are	local OCR	measured boxes; a model drifted 12–15 pt between two reads of the same table. Also the last tier — no account, no key, still runs

Then the document is checked against itself, so correctness is decidable without a human:

IDENTITY	HERE
<code>line_net == qty × unit_net × (1 – disc)</code>	450.00 , 120.00
<code>net == ∑ line_net · vat == ∑ (line × vat%) · gross == net + vat</code>	570.00 · 108.30 · 678.30

A break means a misread digit: re-prompt once, then halt. Money is `Decimal`, never `float`. Honest limit — a misread SKU passes all four, caught later by Fakturama's own selector finding no match.

Verifying, and stopping

INVARIANT	MEANING
Selection is the existence check	never query the database — ask the Order's own picker. Not selectable <i>from the Order</i> = correctly absent
Saving is confirmed by re-selection	the proof a Debtor persisted is that the Order's picker now finds it
Ambiguity halts	>1 exact candidate, or one conflicting with the source → one <code>AmbiguityHalt</code> with the control, query, candidates and a screenshot; non-zero exit

Guessing between two Debtors is the one failure this must never produce. Idempotency falls out: every stage is `check` → `create` → `re-select`.

What it costs

DECISION	PAID
UIA first, vision only as a tiebreak	Windows-only; SWT's thin <code>AutomationId</code> coverage is what forces T2
Arithmetic self-check over a second OCR pass	blind to errors that preserve the identities
Halt on ambiguity	the interesting cases need a human